

ATHARVA WAGH

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SUMMARY

Software Engineer & Computer Science graduate student with 2+ years of experience building scalable distributed systems, backend services, and cloud-native applications. Strong foundation in data structures, algorithms, and system design, with hands-on experience developing high-performance, containerized systems in Linux environments. Proficient in Python, Java, and C#, with experience in asynchronous processing, distributed computing, and API development. Passionate about solving complex engineering problems and building reliable systems at scale.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C#, SQL, C++,

AI/ML: LLMs, Retrieval-Augmented Generation (RAG), Model Evaluation, Scikit-learn, TensorFlow, PyTorch, Tesseract-OCR.

Automation & Integration: REST APIs, JSON, Workflow Automation, System Integration

Distributed & Backend Systems: Microservices, Distributed Systems, Asynchronous Programming, OOP, System Design, Data Pipelines, Parallel Processing, Rabbit MQ, Apache Kafka

Frameworks & APIs: .NET, Unity, Next.js, VueJS, React, Angular, Node.js, Django, Springboot, Jira

Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD, Git, Kafka, Docker Compose

Databases: PostgreSQL, MongoDB, MySQL, Cassandra, Firebase

Data & Analytics: SQL, Pandas, NumPy, Power BI, Data Visualization, ETL Pipelines

Core Competencies: Data Structures & Algorithms, Operating Systems, Computer Architecture, Distributed Systems, Process Automation, Problem Solving, Data Analysis, Debugging, Cross-functional Collaboration

EDUCATION

•University of Southern California, Los Angeles, CA

GPA: 3.3/4.0

Master of Science in Computer Science

Aug 2025 - May 2027

– Relevant Coursework: Analysis of Algorithms, High-Performance Computing, Database Systems, Distributed Systems, Advanced Computer Vision, Deep Learning, Data Science for Machine Learning and applications.

•University of Mumbai, Mumbai, India

GPA: 4.0/4.0

Bachelor of Technology in Information Technology (Blockchain Honors)

Aug 2020 - May 2024

– Magna Cum Laude Equivalent | Key Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Cloud Computing, Computer Organization and Architecture, IoT.

PROFESSIONAL EXPERIENCE

•Bioinformatics Data Analyst (Research Assistant)

Los Angeles, CA

University of Southern California

Sep 2025 - Present

- Developed automated data pipelines to process 100M+ RNA sequencing reads, improving data processing reliability and reproducibility.
- Designed structured workflows for handling large datasets, ensuring clear documentation of processing steps and system outputs.
- Performed comparative analysis and validation of bioinformatics tools to evaluate data integrity and pipeline performance.
- Produced reproducible analysis scripts and documentation to support collaborative research and review processes.
- Documented and extracted Receptor Response and presented the Statistical Analysis using Microsoft Excel

•Software Developer [Full Time]

Mumbai, India

Institute of Management and Foreign Studies (IMFS)

Jul 2024 - May 2025

- Designed and deployed AI-powered automation modules using Python and AWS SageMaker to streamline CRM workflows.
- Built automated data pipelines and backend services for data extraction, and transformation across enterprise systems.
- Developed API-based integrations enabling seamless communication between services using structured JSON data.
- Implemented containerized workflows (Docker, Kubernetes) improving scalability and reducing processing time by 83%.
- Collaborated with stakeholders to translate business requirements into technical automation solutions, improving operational efficiency.
- Implemented ONNX cross platform machine learning model deployment to develop models in PyTorch and deploy in .NET MVC for scalability and production performance.

•Web Developer Intern [Co-Op]

Mumbai, India

Maabella SkinCare

Jun 2023 - May 2024

- Engineered a full-stack CRM platform supporting 1000+ requests/hour, improving business process efficiency through automation.
- Built RESTful APIs and database-driven workflows to automate data handling and reporting processes.
- Designed real-time analytics dashboards and reporting pipelines, enabling data-driven decision-making.
- Performed testing, debugging, and iterative improvements to enhance reliability and user experience.
- Developed OCR-based product recommendation feature using FastAPI and Docker, improving user engagement.

ACADEMIC PROJECTS

LLM Powered Interview Officer

2024

LLM Agents | LangChain | Web Scraping | RAG | Docker | WebSockets | NLP | Apache Kafka

- Built an AI-driven automation system using Retrieval-Augmented Generation (RAG) for intelligent question-answering and decision support.
- Designed automated workflows integrating LLMs with external data sources, enabling contextual responses.
- Implemented real-time, event-driven communication using WebSockets and distributed messaging (Kafka).
- Deployed containerized services, simulating scalable enterprise automation systems.

TCR Sequence Extraction from RNA-seq Data

2025

Bioinformatics Data Pipeline | Python | R | RStudio | Parallel Processing

- Developed a high-performance data pipeline processing 100M+ records, using parallelization and chunking strategies.
- Optimized compute efficiency and throughput, achieving 76% alignment performance.
- Designed reproducible workflows and automated scripts to ensure reliability and maintainability.
- Applied systematic testing and benchmarking methodologies to validate pipeline performance.

Predictive Analytics for College Admissions Application

2023

Machine Learning Project | Python | TypeScript | AWS Sagemaker | NumPy | Sci-Kit Learn | Node.js | Seaborn

- Collected application data; performed EDA, handled outliers, and applied mean-based imputation; built ensemble pipelines with Random Forest and XGBoost for multi-class classification
- Achieved ROC AUC scores of 0.81 (train) and 0.78 (test); deployed model via FastAPI with REST endpoints for real-time predictions
- Implemented cloud-based deployment workflows for scalable model inference.

RESEARCH EXPERIENCE

Bridging Energy Gaps: Blockchain-Enabled P2P Trading for Renewable Energy

2024

- Authored peer-reviewed research on decentralized energy trading systems with distributed algorithms and smart contracts. Developed optimization algorithms for energy distribution system architecture
- Developed optimization algorithms for real-time IoT sensor integration and automated distributed system architectures